

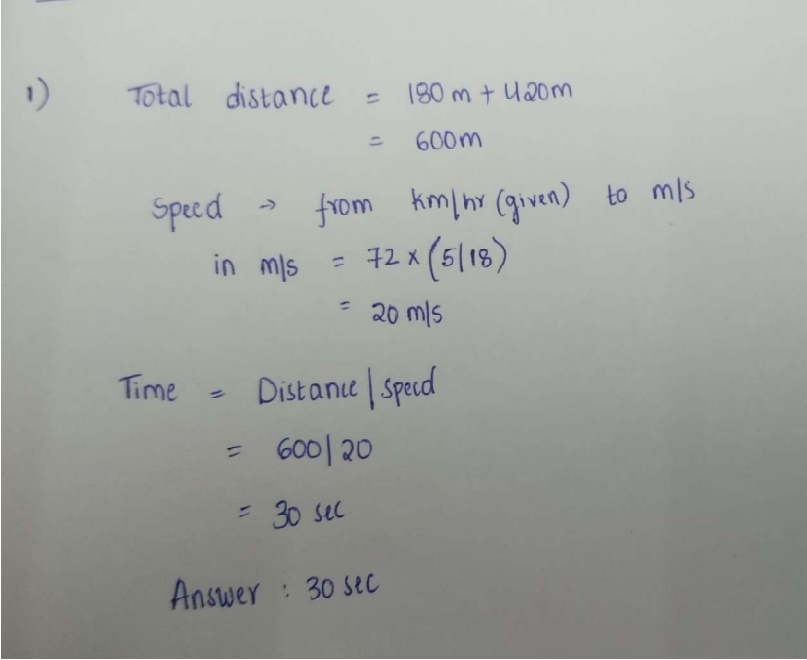
QRA Assignment

S20230010045

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Question 1: Time, Speed & Distance

A 180-meter long train is running at 72 km/hr. It approaches a bridge which is 420 meters long. How much time will the train take to cross the bridge completely?



Handwritten solution for Question 1:

1) Total distance = 180 m + 420 m
= 600 m

Speed → from km/hr (given) to m/s
in m/s = $72 \times \left(\frac{5}{18}\right)$
= 20 m/s

Time = Distance / speed
= $600 / 20$
= 30 sec

Answer : 30 sec

Question 2: Time & Work

P and Q can build a wall in 15 days. Q and R can build the same wall in 18 days. R and P can build it in 24 days. How many days will it take for P, Q, and R working together to complete the wall?

2) (P, Q) 1-day work = $\frac{1}{15}$
(Q, R) 1-day work = $\frac{1}{18}$
(R, P) 1-day work = $\frac{1}{24}$

now combine work (P+Q+R)

$$2(P, Q, R) = \left(\frac{1}{15} + \frac{1}{18} + \frac{1}{24}\right)$$
$$(P, Q, R) = \left(\frac{1}{15} + \frac{1}{18} + \frac{1}{24}\right) \times \frac{1}{2}$$

LCM of 15, 18, 24 = 360

$$(P, Q, R) = \frac{59}{720}$$

Total time = $\frac{1}{\text{work per day}}$

$$= \frac{720}{59} = 12.2 \text{ days}$$

Answer = 12.2 days or $\frac{720}{59}$ days

Question 3: Percentages

In a college election, the winning candidate received 60% of the total valid votes. 8% of the votes were invalid. If the total number of votes cast was 85,000, how many valid votes did the losing candidate receive?

3) Total votes = 85,000
Invalid votes = 8 %
Total Valid Votes
= 85,000 - 8 % of 85,000
= 92 % of 85,000
= 78,200
Winner gets 60 % Valid votes
Loser gets 40 % Valid votes
Loser votes = 40 % of 78,200
= 31,280
Answer: 31,280 votes

Question 4: Profit, Loss & Discounts

A dealer marks the price of an item 50% above its cost price. He then sells it at two successive discounts of 15% and 10%. What is his net profit or loss percentage?

u) Assume cost price (CP) be 100 ₹

$$\begin{aligned}\text{Marked price (MP)} &= CP + 50\% \\ &= 100 + 50 \\ &= 150\end{aligned}$$

$$\begin{aligned}\text{First discount} &= \text{price after } 15\% \text{ discount} \\ &= 150 - (0.15 \times 150) \\ &= 150 - 22.5 = 127.5\end{aligned}$$

$$\begin{aligned}\text{Second discount} &= \text{selling price (SP)} \\ &= 127.5 - (0.10 \times 127.5) \\ &= 114.75\end{aligned}$$

$$\begin{aligned}\text{Profit \%} \\ \text{Profit} &= SP - CP = 114.75 - 100 \\ &= 14.75\end{aligned}$$

$$\begin{aligned}\text{Profit \%} &= \left(\frac{14.75}{100} \right) \times 100 \\ &= 14.75\%\end{aligned}$$

Answer = Profit of 14.75%

Question 5: Ratios & Proportions

The ratio of apples to oranges in a basket is 11:9. If 30 apples are added and 10 oranges are removed, the new ratio becomes 4:3. What was the original total number of fruits in the basket?

5)

let Original apples = $11x$

Original oranges = $9x$

New apples = $11x + 30$

New oranges = $9x - 10$

new ratio is 4:3

$$\frac{11x+30}{9x-10} \approx \frac{4}{3}$$

$$3(11x+30) = 4(9x-10)$$

$$x = 130/3$$

Original total fruits

$$= 11x + 9x$$

$$= 11\left(\frac{130}{3}\right) + 9\left(\frac{130}{3}\right)$$

$$= 20 \times \frac{130}{3}$$

$$= \frac{2600}{3} \approx 867$$

$$\underline{\text{Final Answer}} = \frac{2600}{3} \approx 867$$

Question 6: Mixtures & Solutions

A 60-liter mixture of soda and water contains them in the ratio 7:5. How many liters of soda must be added to make the ratio 5:3?

6)

$$\text{Total parts} = 7+5 = 12$$

$$\text{Soda} = \left(\frac{7}{12}\right) \times 60 = 35 \text{ L}$$

$$\text{Water} = \left(\frac{5}{12}\right) \times 60 = 25 \text{ L}$$

Let Soda added be x liters.

Water remains constant at 25 L

Given New ratio is 5:3

$$\frac{35+x}{25} = \frac{5}{3}$$

$$3(35+x) = 25 \times 5$$

$$105 + 3x = 125$$

$$3x = 20$$

$$x = 6.67 \text{ L}$$

Final answer : $\frac{20}{3}$ liters ≈ 6.67

Question 7: Averages

The average salary of 30 employees in a company is ₹45,000. The average salary of the first 15 employees is ₹42,000 and the average salary of the last 14 employees is ₹48,000. What is the salary of the 16th employee?

$$7) \text{ Total salary} = \text{Total for 30 employees}$$

$$= 30 * 45,000 = 1,350,000$$

$$\text{Sum of first 15: sum} = 15 * 42,000$$

$$= 630,000$$

$$\text{Sum of first 14} = 14 * 48,000$$

$$= 672,000$$

$$\text{Salary of 16}^{\text{th}} = \text{Total} - \left(\begin{array}{l} \text{sum first} \\ \text{salary of 15}^{\text{th}} + \\ \text{sum of first 14} \end{array} \right)$$

$$= 1,350,000 - (630,000 + 672,000)$$

$$= 48,000$$

Answer : 48,000

Question 8: Partnerships

A and B start a business with investments of ₹80,000 and ₹1,20,000 respectively. After 6 months, C joins them with an investment of ₹1,50,000. If the annual profit is ₹2,07,000, what is B's share?

8)

9)

$$\begin{aligned} \text{A's Share} &= 80,000 \times 12 \text{ months} \\ &= 9,60,000 \\ \text{B's Share} &= 1,20,000 \times 12 \text{ months} \\ &= 14,40,000 \\ \text{C's Share} &= 1,50,000 \times 6 \text{ months} \\ &= 9,00,000 \end{aligned}$$

Profit sharing ratio

$$\begin{aligned} \text{A:B:C} &= 9,60,000 : 14,40,000 : 9,00,000 \\ &= 16 : 24 : 15 \end{aligned}$$

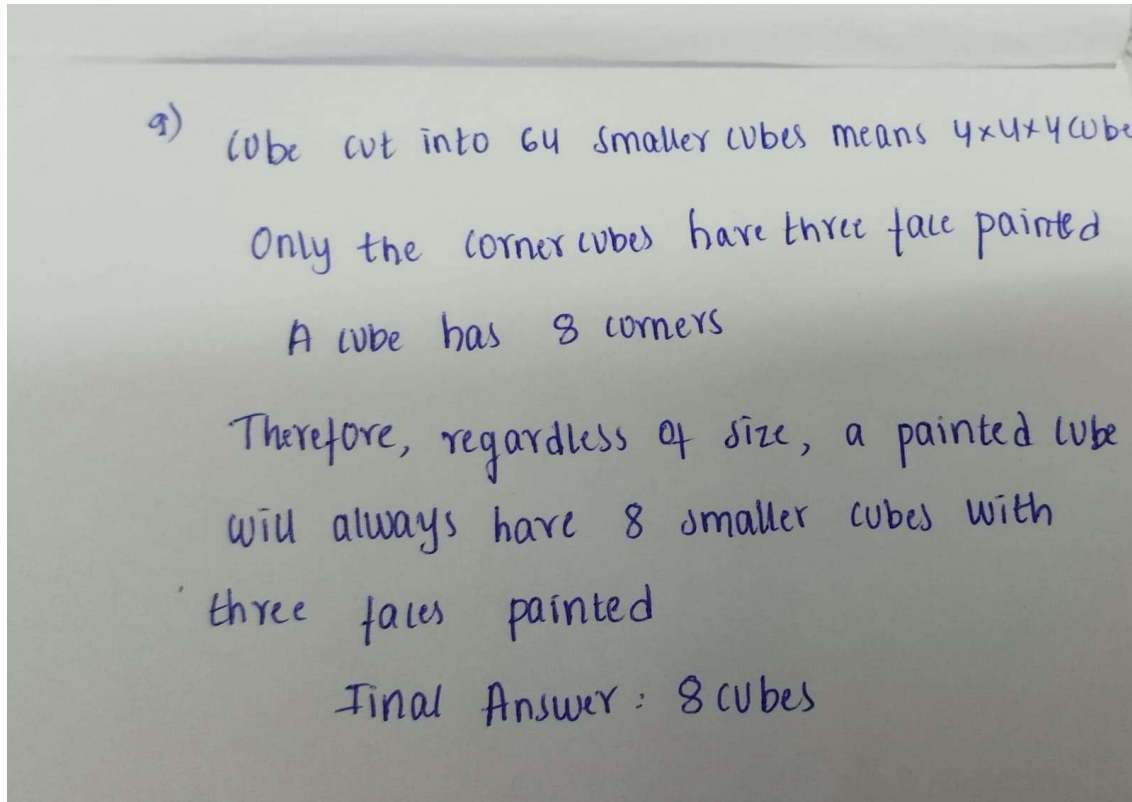
B's share.

$$\begin{aligned} \text{Total parts} &= 16 + 24 + 15 = 55 \\ \text{B's Share} &= \frac{24}{55} \times 2,07,000 \\ &= 90,327.27 \end{aligned}$$

Final Answer: 90,327.27

Question 9: Cubes, Cuboids & Dices

A large cube is painted red on all faces and cut into 64 identical smaller cubes. How many of these smaller cubes will have exactly three faces painted red?



Question 10: Mensuration & Geometry

A right-angled triangle is inscribed in a semicircle such that its hypotenuse is the diameter. If the other two sides measure 9 cm and 12 cm, what is the circumference of the semicircle?

10) Given sides 9, 12.

$$\text{hypotenuse}^2 = 9^2 + 12^2 = 225 \text{ (using pythagoras)}$$

$$\text{hypotenuse} = 15 \text{ cm}$$

$$\text{Diameter of semi circle} = 15 \text{ cm}$$

$$\text{Radius} = 15/2 = 7.5 \text{ cm}$$

$$\text{Circumference of full circle} = 2\pi r$$

$$\text{Circumference of semicircle} = \pi r + \text{diameter}$$

$$= \pi (7.5) + 15 = 7.5\pi + 15 \text{ cm}$$

$$\text{Final : } (7.5\pi + 15) \text{ cm.}$$